1)

```
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine start alpha ./rootfs-alpha /bin/sh --soft-mib 48
--hard-mib 80
OK: container 'alpha' started, pid=4604
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine start beta  ./rootfs-beta  /bin/sh --soft-mib 64
--hard-mib 96
OK: container 'beta' started, pid=4613
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ ps aux | grep -E "engine|alpha|beta" | grep -v grep
samiksha    2367  0.0  0.1 245444  7668 ?        Sl   Apr06   0:00 /usr/libexec/ibus-engine-simple
root        4584  0.0  0.1  28356  7876 pts/1    S+   00:11   0:00 sudo ./engine supervisor ./rootfs-base
root        4585  0.0  0.0  28356  2640 pts/2    Ss   00:11   0:00 sudo ./engine supervisor ./rootfs-base
root        4586  0.0  0.0  28364  1772 pts/2    Sl+  00:11   0:00 ./engine supervisor ./rootfs-base
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$
```

```
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ cd ~/OS-Jackfruit/boilerplate
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine supervisor ./rootfs-base
[sudo] password for samiksha:
Supervisor started. base-rootfs=./rootfs-base  socket=/tmp/mini_runtime.sock
```

2)

```
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
ID              PID      STATE      SOFT(MiB)  HARD(MiB)    EXIT       LOG
beta            4613     running    64         96           0          logs/beta.log
alpha           4604     running    48         80           0          logs/alpha.log
```

3)

```
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine stop alpha
OK: sent SIGTERM+SIGKILL to 'alpha' (pid=4604)
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ cp -a ./rootfs-base ./rootfs-alpha2
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ cp cpu_hog ./rootfs-alpha2/
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine run alpha2 ./rootfs-alpha2 "/cpu_hog 10" &
[1] 4851
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ OK: container 'alpha2' started, pid=4854

samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ DONE exit_code=0
0
0: command not found
[1]+  Done                    sudo ./engine run alpha2 ./rootfs-alpha2 "/cpu_hog 10"
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ sleep 3
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ cat logs/alpha2.log
cpu_hog alive elapsed=1 accumulator=6501497280510727664
cpu_hog alive elapsed=2 accumulator=1636784352611753594
cpu_hog alive elapsed=3 accumulator=11912204811354337068
cpu_hog alive elapsed=4 accumulator=2772935416484976851
cpu_hog alive elapsed=5 accumulator=17646321540123666994
cpu_hog alive elapsed=6 accumulator=16145959645437827611
cpu_hog alive elapsed=7 accumulator=4525324235767208487
cpu_hog alive elapsed=8 accumulator=16905507523944914807
cpu_hog alive elapsed=9 accumulator=16034651737838250710
cpu_hog done duration=10 accumulator=3757798097079467609
```

4)

```
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine start beta2 ./rootfs-beta "/cpu_hog 30" --soft-mi
b 40 --hard-mib 60
OK: container 'beta2' started, pid=4920
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
ID              PID     STATE     SOFT(MiB)  HARD(MiB)    EXIT       LOG
beta2           4920    running   40         60           0          logs/beta2.log
alpha2          4854    exited    40         64           0          logs/alpha2.log
beta            4613    running   64         96           0          logs/beta.log
alpha           4604    stopped   48         80           0          logs/alpha.log

samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$
```

```
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ cd ~/OS-Jackfruit/boilerplate
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine supervisor ./rootfs-base
[sudo] password for samiksha:
Supervisor started. base-rootfs=./rootfs-base  socket=/tmp/mini_runtime.sock
```

5)

```
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ cp -a ./rootfs-base ./rootfs-memtest
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ cp memory_hog ./rootfs-memtest/
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine run memtest ./rootfs-memtest "/memory_hog 8 1000"
 --soft-mib 20 --hard-mib 80 &
[1] 5011
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ OK: container 'memtest' started, pid=5014
DONE exit_code=0

[1]+  Done                    sudo ./engine run memtest ./rootfs-memtest "/memory_hog 8 1000" --soft-mib 20 --hard-mib 8
0
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ dmesg | grep -E "SOFT|HARD|container_monitor" | tail -20
dmesg: read kernel buffer failed: Operation not permitted
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo dmesg | grep -E "SOFT|HARD|container_monitor" | tail -20
[    2.106634] ata3.00: ATA-6: VBOX HARDDISK, 1.0, max UDMA/133
[    2.108180] scsi 2:0:0:0: Direct-Access     ATA      VBOX HARDDISK    1.0  PQ: 0 ANSI: 5
[  483.069652] [container_monitor] Module loaded. Device: /dev/container_monitor
[ 1249.998458] [container_monitor] Registering container=alpha pid=4604 soft=50331648 hard=83886080
[ 1262.560687] [container_monitor] Registering container=beta pid=4613 soft=67108864 hard=100663296
[ 1534.216081] [container_monitor] Unregister request container=alpha pid=4604
[ 1582.122459] [container_monitor] Registering container=alpha2 pid=4854 soft=41943040 hard=67108864
[ 1592.103141] [container_monitor] Unregister request container=alpha2 pid=4854
[ 1758.397383] [container_monitor] Registering container=beta2 pid=4920 soft=41943040 hard=62914560
[ 1762.228854] [container_monitor] Removing stale entry container=beta pid=4613
[ 1762.228862] [container_monitor] Removing stale entry container=beta2 pid=4920
[ 1788.229012] [container_monitor] Unregister request container=beta2 pid=4920
[ 2003.505570] [container_monitor] Registering container=memtest pid=5014 soft=20971520 hard=83886080
[ 2005.613823] [container_monitor] SOFT LIMIT container=memtest pid=5014 rss=25763840 limit=20971520
[ 2012.781535] [container_monitor] HARD LIMIT container=memtest pid=5014 rss=84484096 limit=83886080
[ 2012.788466] [container_monitor] Unregister request container=memtest pid=5014
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$
```

6)

```
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine stop memtest
OK: sent SIGTERM+SIGKILL to 'memtest' (pid=5014)
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ cp -a ./rootfs-base ./rootfs-memtest2
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ cp memory_hog ./rootfs-memtest2/
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine run memtest2 ./rootfs-memtest2 "/memory_hog 8 500
" --soft-mib 20 --hard-mib 32
OK: container 'memtest2' started, pid=5098
DONE exit_code=0
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo dmesg | grep -E "SOFT|HARD" | tail -10
[    2.106634] ata3.00: ATA-6: VBOX HARDDISK, 1.0, max UDMA/133
[    2.108180] scsi 2:0:0:0: Direct-Access     ATA      VBOX HARDDISK    1.0  PQ: 0 ANSI: 5
[ 2005.613823] [container_monitor] SOFT LIMIT container=memtest pid=5014 rss=25763840 limit=20971520
[ 2012.781535] [container_monitor] HARD LIMIT container=memtest pid=5014 rss=84484096 limit=83886080
[ 2239.025090] [container_monitor] HARD LIMIT container=memtest2 pid=5098 rss=34152448 limit=33554432
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
ID               PID      STATE      SOFT(MiB)  HARD(MiB)    EXIT       LOG
memtest2         5098     killed     20         32           0          logs/memtest2.log
memtest          5014     killed     20         80           0          logs/memtest.log
beta2            4920     exited     40         60           0          logs/beta2.log
alpha2           4854     exited     40         64           0          logs/alpha2.log
beta             4613     running    64         96           0          logs/beta
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$
```

7)

```
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ time sudo ./engine run cpuhi ./rootfs-cpuhi "/cpu_hog 20" --nic
e -5
OK: container 'cpuhi' started, pid=5179
DONE exit_code=0

real    0m19.174s
user    0m0.006s
sys     0m0.011s
```

```
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ time sudo ./engine run cpulo ./rootfs-cpulo "/cpu_hog 20" --nic
e 10
[sudo] password for samiksha:
OK: container 'cpulo' started, pid=5206
DONE exit_code=0

real    0m23.220s
user    0m0.006s
sys     0m0.008s
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$
```

8)

```
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ ps aux | grep -E "engine|defunct" | grep -v grep
samiksha    2337  0.0  0.1 245444  7688 ?        Sl   10:22   0:00 /usr/libexec/ibus-engine-simple
root        3906  0.0  0.1  28324  7588 pts/1    S+   10:29   0:00 sudo ./engine supervisor ./rootfs-base
root        3907  0.0  0.0  28324  2604 pts/2    Ss   10:29   0:00 sudo ./engine supervisor ./rootfs-base
root        3908  0.0  0.0  93900  2096 pts/2    Sl+  10:29   0:00 ./engine supervisor ./rootfs-base
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo kill $(pgrep -f "engine supervisor")
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ ps aux | grep -E "engine|defunct" | grep -v grep
samiksha    2337  0.0  0.1 245444  7692 ?        Sl   10:22   0:00 /usr/libexec/ibus-engine-simple
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo dmesg | grep container_monitor | tail -5
[  291.534143] [container_monitor] Unregister request container=beta pid=3832
[  467.796565] [container_monitor] Registering container=alpha pid=3915 soft=50331648 hard=83886080
[  476.337506] [container_monitor] Registering container=beta pid=3923 soft=67108864 hard=100663296
[  485.472065] [container_monitor] Unregister request container=alpha pid=3915
[  495.732034] [container_monitor] Unregister request container=beta pid=3923
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$
```

```
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine supervisor ./rootfs-base
Supervisor started. base-rootfs=./rootfs-base  socket=/tmp/mini_runtime.sock
Supervisor shutting down...
Supervisor exited cleanly.
samiksha@samiksha-VirtualBox:~/OS-Jackfruit/boilerplate$
```